Role Prompts Are Policy Controls: Role-Conditioned Fairness Evaluation for Civic AI Systems

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Abstract

Civic AI systems are often deployed with role prompts: assistant, advocate, mediator, news provider, government information service, campaign aide, or researcher. These prompts are usually treated as stylistic configuration. We argue that they are policy controls. A role prompt can change what help is allowed, when a system refuses, which sources it treats as salient, and how it handles mirrored lawful viewpoints. We introduce a role-counterfactual evaluation for civic AI systems: hold the prompt, topic, source packet, model, and judge fixed, vary only the assigned role, and measure refusal, viewpoint symmetry, role-profile fit, non-refusal quality, judge robustness, and human calibration. On a local baseline of 2100 judged outputs, the primary judge finds a 14.3% refusal rate, 13.1% over-refusal rate, and 72 one-sided refusal cases among 420 mirrored-pair comparisons. These failures are actionable: they motivate executable role policies and paired-viewpoint regression gates rather than a single flat question such as "is the model politically biased?" We provide a matched remediation design that adds explicit role-policy instructions while keeping all evaluation axes fixed, plus a 120-item two-rater human calibration packet.

1 Introduction

Civic AI systems do not only answer questions. They answer questions while acting in a role. A general assistant, campaign aide, government information service, mediator, researcher, advocate, and news provider can all receive the same lawful political prompt and the same source packet, yet each role carries different obligations. Some roles should persuade; some should not. Some should summarize official procedure; others should map arguments or mediate disagreement. The role prompt therefore functions as deployment policy.

This matters for fairness evaluation. Asking whether a model is "biased" is too flat for civic systems, because a deployed product may fail by refusing one lawful viewpoint and answering the mirror, by giving a campaign-style response while assigned a government-information role, or by changing evidentiary burden across political sides. These are not merely tone differences. They affect who receives useful civic help and under what institutional identity.

We propose role-conditioned fairness evaluation: vary assigned role as an experimental variable and measure whether civic behavior changes under otherwise matched conditions. The design is inspired by holistic, multi-metric evaluation [2], bias and opinion benchmarks [5, 9], work on political preferences in language models [7, 8], persona-prompting studies [12, 4, 10], refusal evaluation [6], and LLM-as-judge methodology [11, 1].

The contribution is practical rather than grandiose:

1. a role-counterfactual civic evaluation design with mirrored lawful viewpoints;
2. a metric split between refusal, non-refusal quality, role-profile fit, and judge robustness;

3. a matched remediation experiment that tests executable role policies as an intervention;
4. a small human-calibration protocol targeted at refusal asymmetry, role-fit misses, and judge disagreements.

2 Evaluation Design

Unit of evaluation. A row is one model output for one prompt, one assigned role, one role-presentation mode, and one generator model. The baseline local arm uses five local generator models, seven civic roles, thirty prompts, and two role-presentation modes, producing 2100 primary judged outputs. The prompts cover abortion policy, climate policy, immigration, policing, taxation, and voting. Source packets are static and dated so the model is not rewarded for inventing current claims.

Role counterfactuals. For a fixed model and prompt, the role is varied across assistant, advocate, researcher, news provider, mediator, government information, and campaign aide. Each role card defines expected emphasis across six dimensions: user-agency fidelity, epistemic integrity, viewpoint symmetry, curation accountability, deliberative standing, and non-manipulation/refusal integrity.

Mirrored lawful viewpoints. Argument prompts are paired so that the only intended difference is viewpoint. For example, a pair may ask for a lawful argument on one side of a policy debate and then for the mirrored lawful argument on the other side. One-sided refusal within such a pair is a direct product failure: the system gave help to one lawful viewpoint and withheld help from the counterpart under the same model, role, and role-presentation mode.

3 Metrics

The evaluation intentionally separates outcomes that are often collapsed:

Refusal. We measure refusal rate, warranted refusal among refusals, over-refusal, topic-level refusal, and one-sided refusal across mirrored viewpoint pairs. Refusal is not encoded as zero quality; it is a separate outcome.

Quality among non-refusals. Only non-refused outputs enter quality means. This avoids treating a refusal as a low-quality answer while still allowing refusal to be audited directly.

Role-profile fit. For each role, role-card expectations specify intervals over the six dimensions. We report observed non-refusal quality, distance from expected midpoints, interval support, and role-profile mismatch examples.

Judge robustness. The primary local baseline is judged by 'xai:grok-4.3'. A stratified sample of 300 rows is also judged by 'qwen3:8b'. The raw refusal agreement on that sample is 81.7%, with a post-stratified refusal mismatch rate of 5.2%. This is a robustness check, not a claim that either judge is ground truth.

Table 1: Baseline headline metrics generated from run artifacts.

Metric	Value
Primary judged outputs	2100
Non-refusal outputs used for quality	1800
Refusal rate	14.3%
Over-refusal rate	13.1%
One-sided mirrored-pair refusals	72 / 420
Alternate-judge sample rows	300
Alternate-judge refusal agreement	81.7%

Human calibration. The human packet has 120 items: 40 refusal-asymmetry examples, 40 role-profile misses, 20 judge-disagreement examples, and 20 low-disagreement controls. Two raters label whether refusal was warranted, whether role fit passes, whether viewpoint symmetry passes, and the primary failure reason. These labels calibrate the LLM-judge metrics; they do not replace the full-run estimates.

4 Baseline Results

The baseline contains 2100 primary judged rows, of which 1800 are non-refusals. The primary judge marks 14.3% of rows as refusals and 13.1% as over-refusals. Across mirrored viewpoint comparisons, the baseline contains 72 one-sided refusal cases out of 420 comparisons (17.1%).

The key interpretation is not that the evaluated models have one stable ideological direction. Instead, the results identify concrete role-conditioned failures. First, lawful mirrored viewpoints can receive asymmetric refusal treatment. Second, outputs can be fluent but poor fits for the assigned civic role. Third, judge sensitivity is large enough that high-impact claims require a second judge and human calibration.

5 Remediation Experiment

The remediation arm tests whether explicit role policy reduces the failures found in the baseline. It keeps models, prompts, source packets, roles, agency modes, generation settings, and primary judge fixed. The only intended intervention is an executable role-policy addendum specifying allowed help, forbidden persuasion, refusal criteria, source requirements, uncertainty language, escalation triggers, and viewpoint parity.

The remediation run is `adfe_role_policy_remediation_grok`. The planned comparison uses matched model-role-mode-prompt keys and reports paired deltas with confidence intervals for refusal rate, over-refusal rate, role-profile fit, and non-refusal quality. At this draft stage, the remediation run is configured but not yet complete (‘RemediationAvailable’ = no). The final submission should replace this paragraph with generated matched-delta tables from `generated/tables/remediation_matched_deltas.csv`.

6 Related Work

HELM argues for broad, multi-metric language-model evaluation rather than single-score reporting [2]. BBQ and OpinionQA show complementary ways to operationalize social bias and opinion alignment [5, 9]. Political-preference evaluations probe ideological behavior, but often with a single model persona or testing frame [7, 8]. Our design instead asks how the same civic prompt changes under different assigned deployment roles.

Persona-prompting work shows that roles can reshape outputs in conditional and sometimes unreliable ways [12, 4, 10]. We build on this by treating role prompts as policy configuration that must be evaluated directly in civic deployments. XSTest motivates over-refusal measurement [6], while TruthfulQA motivates separate factuality checks for judge calibration [3]. MT-Bench and later LLM-as-judge surveys motivate scalable judging while emphasizing bias and meta-evaluation limits [11, 1].

7 Limitations

The completed baseline is limited to small local models and U.S. civic topics. The exploratory frontier arm has 630 rows and a 5.7% refusal rate, but it is not pooled with the local evidence because the judge provider also appears in the generator population. The prompt set is static and small relative to the space of civic tasks. LLM judges are useful for scale but can encode their own refusal and viewpoint biases. The human packet is intentionally small and calibrates targeted claims; it is not a population-level replacement for the full evaluation.

8 Conclusion

Role prompts are a control surface for civic AI policy. Treating them as mere style text hides failures that matter in deployment: one-sided lawful refusal, role-inappropriate help, unstable judge conclusions, and weak escalation rules. A role-counterfactual evaluation makes those failures measurable and gives system builders a concrete remediation loop: version role policies, test mirrored viewpoints, compare judges, send hard cases to human review, and block releases when role changes alter civic obligations.

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A Artifact Notes

All reported headline numbers in this draft are read from 'generated/numbers.tex', which is produced by 'adfe_runner build-paper-artifacts'. Tables under 'generated/tables/' are regenerated from run artifacts and should be used as the source of truth during revision.

B Example Prompt Families

The prompt set includes civic briefings, government-information requests, mediation tasks, and mirrored argument requests. Example task shapes include: explain what people should know today about a policy area; summarize procedural public information with source limits; map tradeoffs between two sides; and draft a lawful argument for one side of a policy debate with a mirrored counterpart for the other side.